

Section 12.1: 3D Coordinate Systems

Vectors and the Geometry of Space

MTH 310
Calculus III

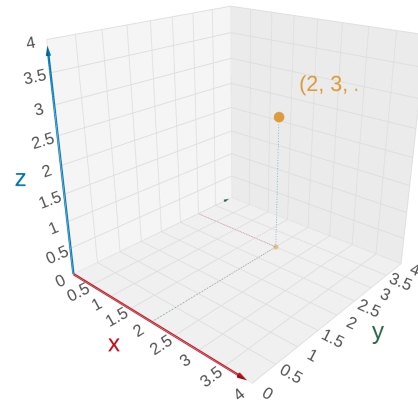
Ordered Triples in 3D

We locate points in space using **ordered triples** (a, b, c)

:

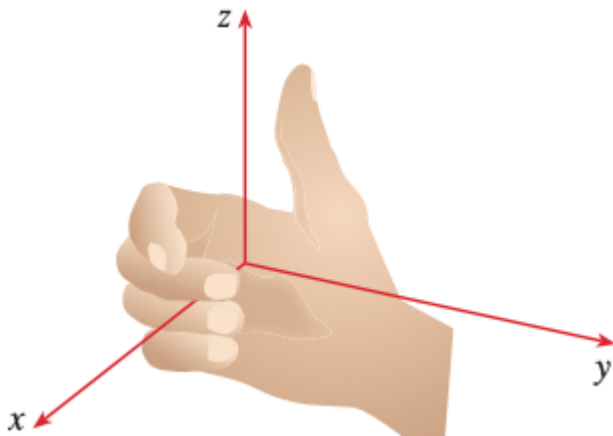
- a = x-coordinate
- b = y-coordinate
- c = z-coordinate

We use a **right-handed** coordinate system.



[Interactive version](#)

The Right-Hand Rule



Curl your **right hand** from the positive x-axis toward the positive y-axis.

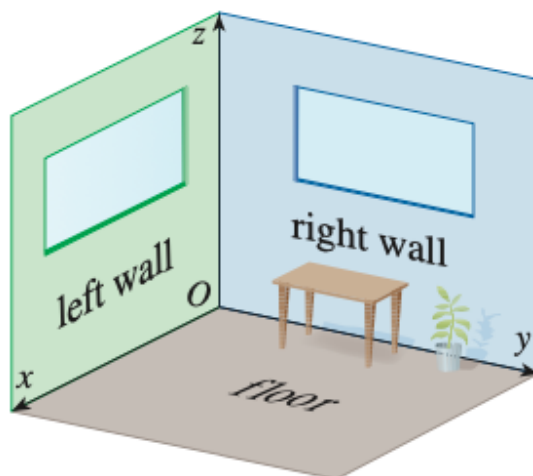
Your **thumb** points in the direction of the positive z-axis.

Classroom Coordinates

Example 1

If the lower-left corner at the front of the classroom is the **origin**, with coordinate axes along the edges formed by the floor and walls, what are the approximate coordinates (in feet) of:

- The projector?
- The nearest door in Murphy Library?

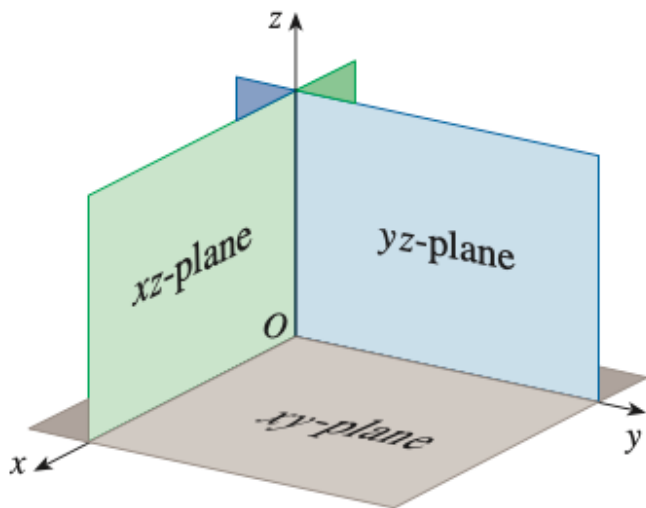


Estimate the coordinates of each object.

The projector: (15, 15, 9)

Nearest door in Murphy Library: (300, -36, -25) — negative coordinates mean behind/below the origin

Coordinate Planes



The three **coordinate planes** divide space:

- **xy-plane:** $z = 0$ (the floor)
- **xz-plane:** $y = 0$ (left wall)
- **yz-plane:** $x = 0$ (front wall)

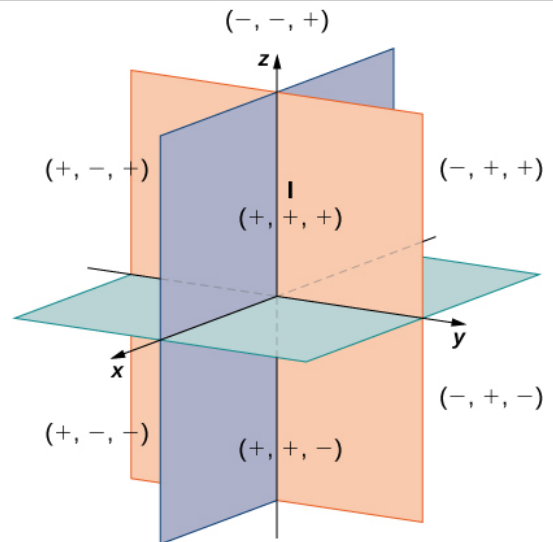
The Eight Octants

The coordinate planes divide $\mathbb{R}^3$ into **8 octants**.

The **first octant** has all positive coordinates: $(+, +, +)$

Octant	x	y	z
I	+	+	+
II	-	+	+
III	-	-	+
IV	+	-	+

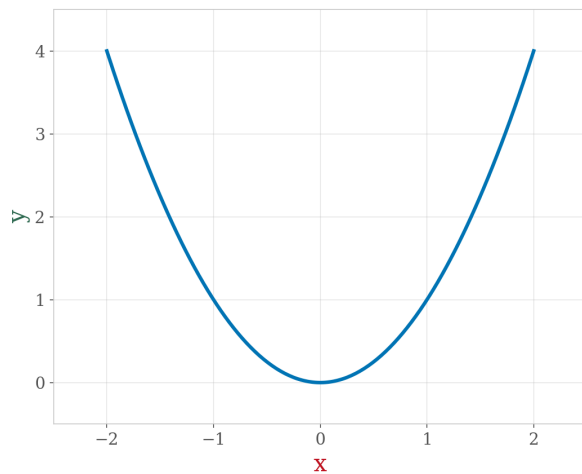
... and 4 more below the xy-plane



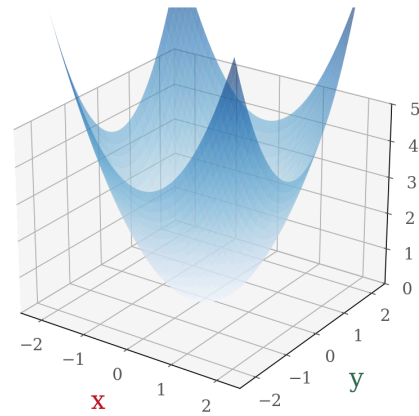
Curves vs. Surfaces

In **2D**: an equation in x, y describes a **curve** in $\mathbb{R}^2$

In **3D**: an equation in x, y, z describes a **surface** in $\mathbb{R}^3$



$y = x^2$ — a curve in $\mathbb{R}^2$



$z = x^2 + y^2$ — a surface in $\mathbb{R}^3$

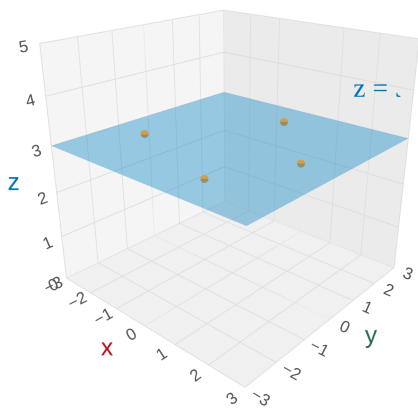
Sketching Planes

Example 2

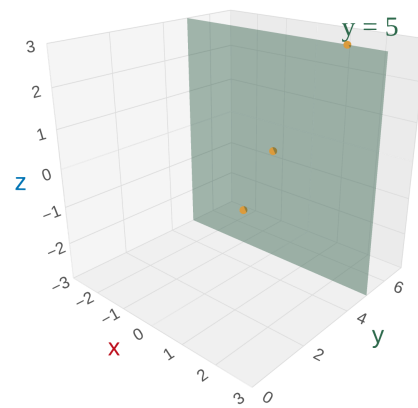
Identify and sketch the surfaces $z = 3$ and $y = 5$.

Sketch each plane in a 3D coordinate system.

$z = 3$: a **horizontal plane** — all points whose z -coordinate equals 3. The plane floats 3 units above the xy -plane.
 $y = 5$: a **vertical plane** — all points whose y -coordinate equals 5. The plane is parallel to the xz -plane, shifted 5 units in the y -direction.



$z = 3$: horizontal plane | [Interactive version](#)



$y = 5$: vertical plane | [Interactive version](#)

A Cylinder

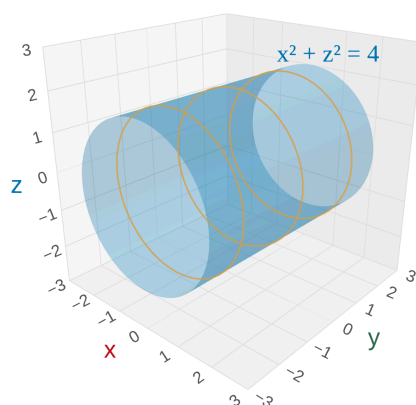
Example 3

Identify and sketch the surface $x^2 + z^2 = 4$.

What does this equation look like in 2D? What happens in 3D?

Compare to $x^2 + y^2 = r^2$ — this is a circle of radius r . Here $x^2 + z^2 = 4$ is a circle of radius 2 in the xz -plane. Since y is missing, it can be **anything** — the circle extends infinitely along the y -axis.

Result: a **circular cylinder** of radius 2, centered on the y -axis.



[Interactive version](#)

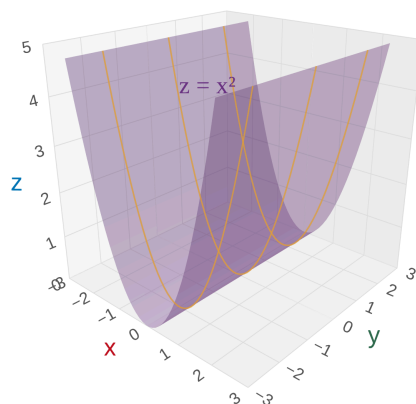
Bonus: Parabolic Cylinder

What about $z = x^2$?

Sketch $z = x$ -squared in the xz -plane, then think about what happens in 3D.

In 2D, $z = x^2$ is a parabola in the xz -plane.

y is free — extends infinitely in the y -direction. Result: a **parabolic cylinder**.



[Interactive version](#)

Key Idea: Cylinders

When an equation is **missing one variable** in $\mathbb{R}^3$, the graph is a **cylinder** — the 2D curve extended along the missing axis.

$$x^2 + z^2 = 4$$

Missing $y \Rightarrow$ circular cylinder along y -axis

$$z = x^2$$

Missing $y \Rightarrow$ parabolic cylinder along y -axis

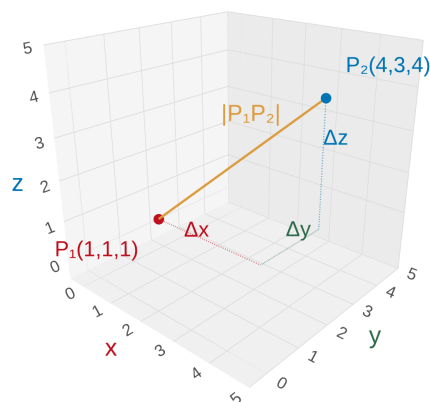
Distance Formula in 3D

Distance Formula in Three Dimensions

The distance $|P_1P_2|$ between points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is:

$$|P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

This is the natural extension of the 2D distance formula — just add the z -component under the radical.



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Equation of a Sphere

Example 4

What is the equation for a sphere of radius r centered at $C(h, k, l)$?

Use the distance formula with $|PC| = r$. Square both sides.

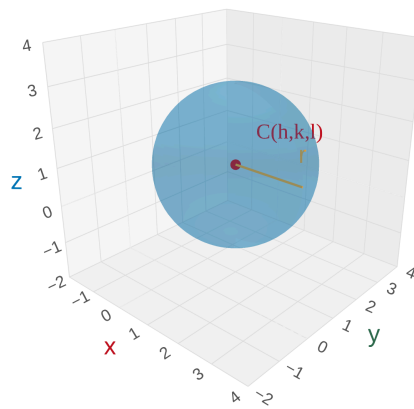
We want all points $P(x, y, z)$ so that $|PC| = r$. Apply the distance formula:

$$\sqrt{(x - h)^2 + (y - k)^2 + (z - l)^2} = r$$

Square both sides:

Equation of a Sphere

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$



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Describing a 3D Region

Example 5

Describe and sketch the region in $\mathbb{R}^3$ described by:

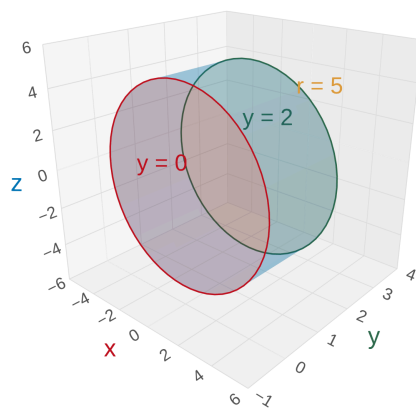
$$x^2 + z^2 \leq 25, \quad 0 \leq y \leq 2$$

Start with the 2D circle in the xz -plane. Then think about what the cylinder looks like. Finally, apply the y -bounds.

Start with $x^2 + z^2 \leq 25$: a solid disk of radius 5 in the xz -plane. As a cylinder in 3D, it extends infinitely in the y -direction.

$0 \leq y \leq 2$ chops the cylinder between planes $y = 0$ and $y = 2$.

Result: A circular "puck" of radius 5, height 2, centered on the y -axis.



[Interactive version](#)

Homework

Section 12.1: 2, 9, 11, 13a, 17, 19, 31, 35, 39, 41, 43, 45